

Module 2

Transient Analysis of First and Second-Order Circuits

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First order circuits

Applying Kirchhoff's laws to purely resistive circuits results in algebraic equations, while applying the laws to RC and RL circuits produces differential equations.

The differential equations resulting from analyzing RC and RL circuits are of the first order.

A first-order circuit is characterized by a first-order differential equation.

There are two types of first-order circuits (RC and RL), and there are two ways to excite the circuits.

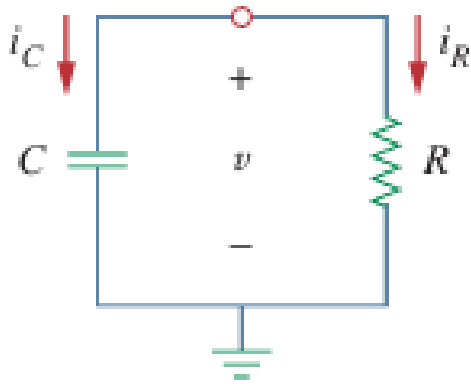
The first way is by initial conditions of the storage elements in the circuits. In these so-called source-free circuits, we assume that energy is initially stored in the capacitive or inductive element. The energy causes current to flow in the circuit and is gradually dissipated in the resistors.

The second way of exciting first-order circuits is by independent sources.

The Source-Free RC Circuit

A source-free RC circuit occurs when its dc source is suddenly disconnected.

The energy already stored in the capacitor is released to the resistors.



Consider a series combination of a resistor and an initially charged capacitor.

Since the capacitor is initially charged, we can assume that at time $t = 0$, the initial voltage is not zero but,

$$v(0) = V_0$$

with the corresponding value of the energy stored as

$$w(0) = \frac{1}{2}CV_0^2$$

The Source-Free RC Circuit

Applying KCL at the top node of the circuit,

$$i_C + i_R = 0$$

By definition, $i_C = C \, dv/dt$ and $i_R = v/R$. Thus,

$$C \frac{dv}{dt} + \frac{v}{R} = 0$$

$$\frac{dv}{dt} + \frac{v}{RC} = 0 \quad \text{This is a first-order differential equation,}$$

To solve this first order differential equation, we rearrange the terms as

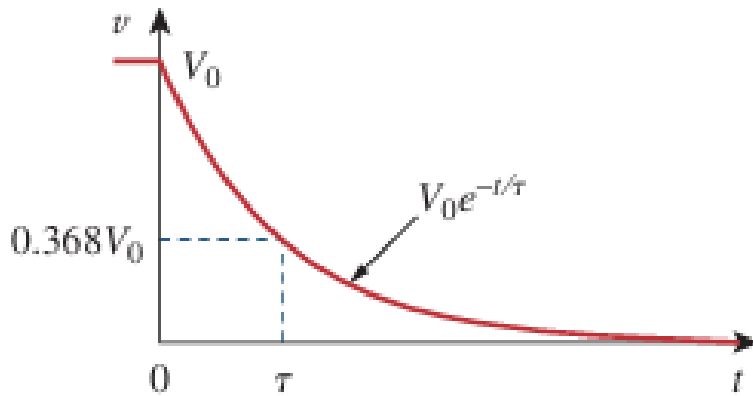
$$\frac{dv}{v} = -\frac{1}{RC} dt$$
$$\ln v = -\frac{t}{RC} + \ln A$$

Taking powers of e produces, $v(t) = Ae^{-t/RC}$

From the initial conditions, $A = V_0$ $v(t) = V_0 e^{-t/RC}$

The Source-Free RC Circuit

The equation $v(t) = V_0 e^{-t/RC}$ shows that the voltage response of the RC circuit is an exponential decay of the initial voltage.



The rapidity with which the voltage decreases is expressed in terms of the time constant, denoted by the lowercase Greek letter tau $\tau = RC$.

Since the response is due to the initial energy stored and the physical characteristics of the circuit and not due to some external voltage or current source, it is called the **natural response** of the circuit.

The natural response of a circuit refers to the behavior (in terms of voltages and currents) of the circuit itself, with no external sources of excitation.

The Source-Free RC Circuit

The time constant of a circuit is the time required for the response to decay to a factor of $1/e$ or 36.8 percent of its initial value.

This implies that at $t = \tau$

$$V_0 e^{-\tau/RC} = V_0 e^{-1} = 0.368 V_0$$

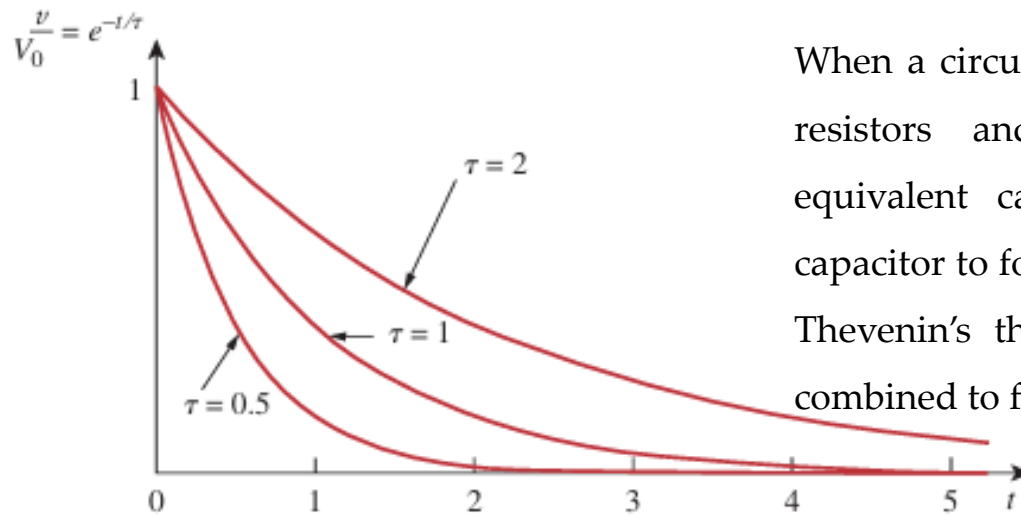
$$\tau = RC$$

In terms of the time constant

$$v(t) = V_0 e^{-t/\tau}$$

$$i_R(t) = \frac{v(t)}{R} = \frac{V_0}{R} e^{-t/\tau}$$

The Source-Free RC Circuit



When a circuit contains a single capacitor and several resistors and dependent sources, the Thevenin equivalent can be found at the terminals of the capacitor to form a simple RC circuit. Also, one can use Thevenin's theorem when several capacitors can be combined to form a single equivalent capacitor.

The power dissipated in the resistor is $p(t) = vi_R = \frac{V_0^2}{R} e^{-2t/\tau}$

The energy absorbed by the resistor up to time t is

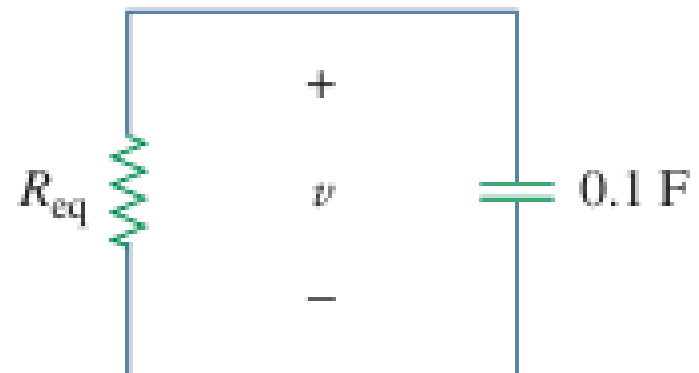
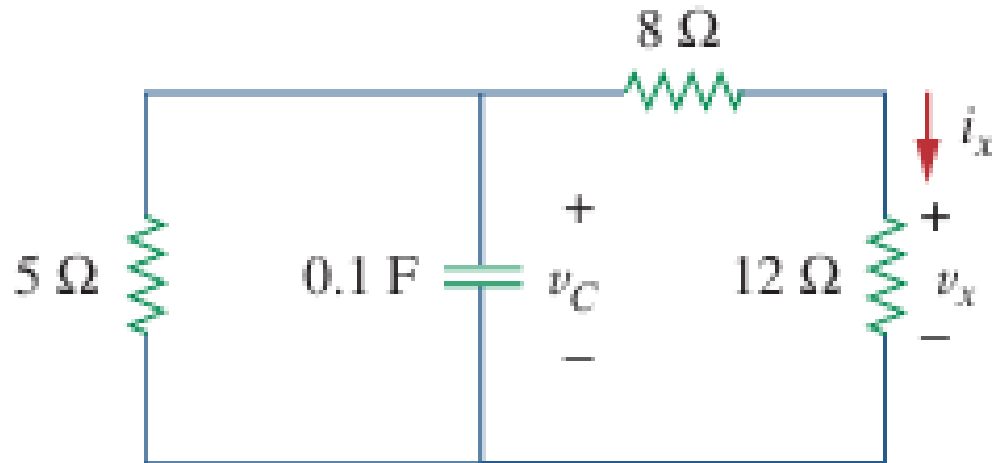
$$\begin{aligned} w_R(t) &= \int_0^t p(\lambda) d\lambda = \int_0^t \frac{V_0^2}{R} e^{-2\lambda/\tau} d\lambda \\ &= -\frac{\tau V_0^2}{2R} e^{-2\lambda/\tau} \Big|_0^t = \frac{1}{2} C V_0^2 (1 - e^{-2t/\tau}), \quad \tau = RC \end{aligned}$$

Notice that as t tends to infinity, w tends to $\frac{1}{2} C V^2$, which is the same as the energy initially stored in the capacitor.

The energy that was initially stored in the capacitor is eventually dissipated in the resistor. ⁷

Exercise

1. Find v_c , v_x and i_x for $t > 0$. Let $v_c(0) = 15$ V.



Exercise

$$R_{\text{eq}} = \frac{20 \times 5}{20 + 5} = 4 \, \Omega$$

$$\tau = R_{\text{eq}}C = 4(0.1) = 0.4 \, \text{s}$$

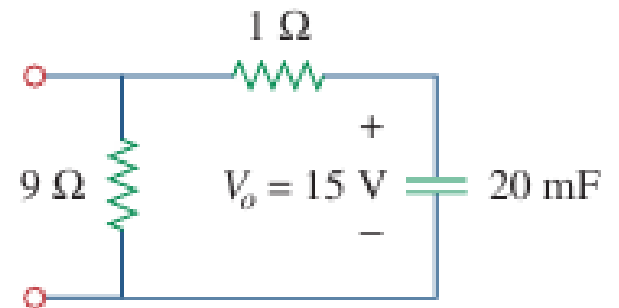
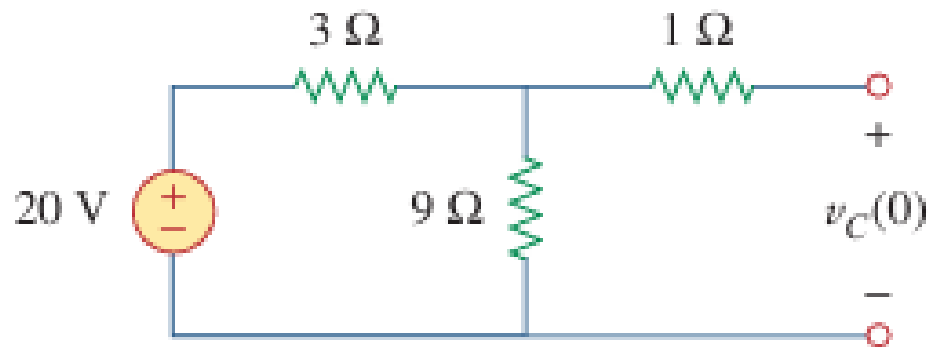
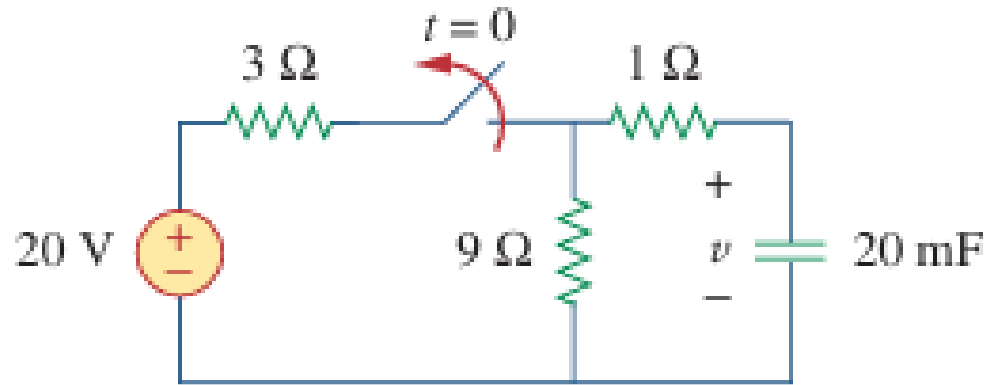
$$v = v(0)e^{-t/\tau} = 15e^{-t/0.4} \, \text{V}, \quad v_C = v = 15e^{-2.5t} \, \text{V}$$

$$v_x = \frac{12}{12 + 8}v = 0.6(15e^{-2.5t}) = 9e^{-2.5t} \, \text{V}$$

$$i_x = \frac{v_x}{12} = 0.75e^{-2.5t} \, \text{A}$$

Exercise

2. The switch in the circuit has been closed for a long time, and it is opened at $t = 0$. Find $v(t)$ for $t \geq 0$. Calculate the initial energy stored in the capacitor.



Exercise

For $t < 0$, the switch is closed; the capacitor is an open circuit to dc,

$$v_C(t) = \frac{9}{9 + 3}(20) = 15 \text{ V}, \quad t < 0$$

$$v_C(0) = V_0 = 15 \text{ V}$$

For $t > 0$, the switch is opened, and we have the RC circuit

$$R_{\text{eq}} = 1 + 9 = 10 \text{ } \Omega$$

$$\tau = R_{\text{eq}}C = 10 \times 20 \times 10^{-3} = 0.2 \text{ s}$$

for $t \geq 0$

$$v(t) = v_C(0)e^{-t/\tau} = 15e^{-t/0.2} \text{ V}$$

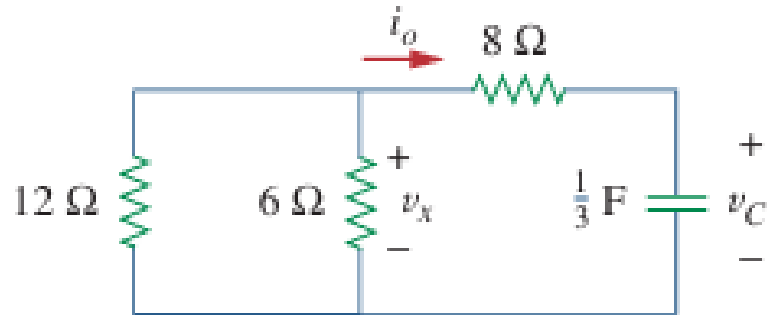
$$v(t) = 15e^{-5t} \text{ V}$$

The initial energy stored in the capacitor is

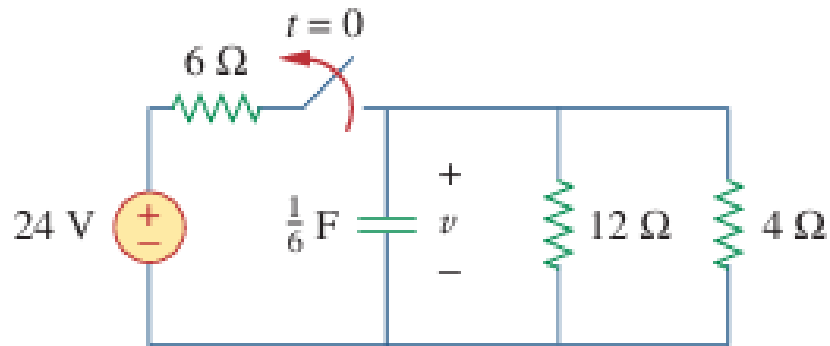
$$w_C(0) = \frac{1}{2}Cv_C^2(0) = \frac{1}{2} \times 20 \times 10^{-3} \times 15^2 = 2.25 \text{ J}$$

Exercise

3. Referring to the circuit below determine v_o , v_x , and i_o for $t \geq 0$. Let $v_c(0) = 60$ V.



4. If the switch opens at $t = 0$, find $v(t)$ for $t \geq 0$ and $w_c(t)$.



Next class: Source-Free RL Circuit